

CYNThERA

Drug Repurposing AI Report

Drug: Doxycycline
Disease: Heart failure
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Recommendation: **PROMISING**

Executive Summary

CYNThERA v2.0 analysis of Doxycycline → Heart failure produced a recommendation of 'PROMISING'. Evidence Strength: 98.7% (HIGH) | Mechanistic Plausibility: 40.9% (LOW) | Risk Level: 5.8% (LOW). 7 claim(s) extracted from literature (7 from openalex), 0 contradiction(s) detected. Prior knowledge: regulatory NONE (0%), repurposing EMERGING, mechanistic MODERATE, clinical HUMAN_EVIDENCE, knowledge maturity SPECULATIVE. Safety grade: B. 5 mechanistic path(s) traced.

Three-Dimensional Score Assessment

Dimension	Score	Level	Details
Support Score (SS)	0.987	HIGH	62 evidence records
Mechanistic Score (MS)	0.409	LOW	10 pathways
Risk Score (RS)	0.058	LOW	0 failed trials, 0 contradictions

Therapeutic Direction Alignment (Phase 4D)

Overall Directional Alignment: **SUPPORTS**

Therapeutic alignment *SUPPORTS* hypothesis: Drug action on primary target(s) (MMP1, MMP13, MMP7, MMP8) is directionally concordant with disease-target therapeutic requirements.

Target	Drug Action	Desired Action	Alignment	Evidence Groups
MMP1	INHIBITION	INHIBITION	SUPPORTS	1 supp / 0 opp groups
MMP13	INHIBITION	INHIBITION	SUPPORTS	1 supp / 0 opp groups
MMP7	INHIBITION	INHIBITION	SUPPORTS	1 supp / 0 opp groups
MMP8	INHIBITION	INHIBITION	SUPPORTS	1 supp / 0 opp groups
RPSB	INHIBITION	UNKNOWN	INSUFFICIENT	0 supp / 0 opp groups

Scientific Context — Prior Knowledge

Dimension	Status	Confidence	Evidence
Regulatory	NONE	0%	No ChEMBL indication match found for this disease
Repurposing	EMERGING	50%	Pair-scoped literature exists for this disease
Mechanistic	MODERATE	41%	Validated path: Doxycycline → MMP1 (P03956) [inhibitor (ChEMBL)] → Ba
Clinical	HUMAN_EVIDENCE	90%	4 registered trial(s) for this pair: NCT03960411 (Phase III, UNKNOWN), NO
Knowledge Maturity	SPECULATIVE	0%	No prior knowledge cache entries for this pair

Candidate Biological Mechanisms

Candidate Mechanism 2: Mechanism 2: MMP1 (P03956) via Basigin interactions (R-HSA-210991)
[WEAK_SPECULATIVE] [DIR: UNKNOWN]

Drug: Doxycycline → Target: MMP1 (P03956) → Reaction: Basigin binds Matrix metalloproteinase-1 (R-HSA-375135) → Pathway: Basigin interactions (R-HSA-210991) → Gene: ATP1B1 → Disease: Heart failure

- Drug: Doxycycline —INHIBITOR→ Target: MMP1 (P03956) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)
Status: DATABASE SUPPORTED; Directionality: SUPPORTED; Source: ChEMBL
- Target: MMP1 (P03956) —COMPLEX_COMPONENT_OF→ Reaction: Basigin binds Matrix metalloproteinase-1 (R-HSA-375135) [\[Open Reactome Reaction\]](#)
Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome
- Reaction: Basigin binds Matrix metalloproteinase-1 (R-HSA-375135) —PART_OF→ Pathway: Basigin interactions (R-HSA-210991) [\[Open Reactome Pathway\]](#)
Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome
- Pathway: Basigin interactions (R-HSA-210991) —CONTAINS_ASSOCIATED_GENE→ Gene: ATP1B1 [\[Open Reactome Pathway\]](#) [\[Open Targets Target\]](#)
Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome + Open Targets/DisGeNET
- Gene: ATP1B1 —ASSOCIATED_WITH→ Disease: Heart failure [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)
Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.

No mapped literature claim supports a critical biological bridge; score is capped below moderate support.

Missing evidence: Direct biological bridge for Target: MMP1 (P03956) → Reaction: Basigin binds Matrix metalloproteinase-1 (R-HSA-375135) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Reaction: Basigin binds Matrix metalloproteinase-1 (R-HSA-375135) → Pathway: Basigin interactions (R-HSA-210991) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Pathway: Basigin interactions (R-HSA-210991) → Gene: ATP1B1 is not supported by a mapped claim.

Candidate Mechanism 1: Mechanism 1: MMP1 (P03956) via Basigin interactions (R-HSA-210991)
[WEAK_SPECULATIVE] [DIR: UNKNOWN]

Drug: Doxycycline → Target: MMP1 (P03956) → Pathway: Basigin interactions (R-HSA-210991) → Gene: ATP1B1 → Disease: Heart failure

- Drug: Doxycycline —**INHIBITOR**→ Target: MMP1 (P03956) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)

Status: DATABASE SUPPORTED; Directionality: SUPPORTED; Source: ChEMBL

- Target: MMP1 (P03956) —**PARTICIPATES_IN**→ Pathway: Basigin interactions (R-HSA-210991) [\[Open Reactome Pathway\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome

- Pathway: Basigin interactions (R-HSA-210991) —**CONTAINS_ASSOCIATED_GENE**→ Gene: ATP1B1 [\[Open Reactome Pathway\]](#) [\[Open Targets Target\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome + Open Targets/DisGeNET

- Gene: ATP1B1 —**ASSOCIATED_WITH**→ Disease: Heart failure [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)

Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.

No mapped literature claim supports a critical biological bridge; score is capped below moderate support.

Missing evidence: Direct biological bridge for Target: MMP1 (P03956) → Pathway: Basigin interactions (R-HSA-210991) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Pathway: Basigin interactions (R-HSA-210991) → Gene: ATP1B1 is not supported by a mapped claim.

Candidate Mechanism 3: Mechanism 3: MMP8 (P22894) via Neutrophil degranulation (R-HSA-6798695) [WEAK_SPECULATIVE] [DIR: UNKNOWN]

Drug: Doxycycline → Target: MMP8 (P22894) → Pathway: Neutrophil degranulation (R-HSA-6798695) → Gene: MME → Disease: Heart failure

- Drug: Doxycycline —**INHIBITOR**→ Target: MMP8 (P22894) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)

Status: DATABASE SUPPORTED; Directionality: SUPPORTED; Source: ChEMBL

- Target: MMP8 (P22894) —**PARTICIPATES_IN**→ Pathway: Neutrophil degranulation (R-HSA-6798695) [\[Open Reactome Pathway\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome

- Pathway: Neutrophil degranulation (R-HSA-6798695) —**CONTAINS_ASSOCIATED_GENE**→ Gene: MME [\[Open Reactome Pathway\]](#) [\[Open Targets Target\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome + Open Targets/DisGeNET

- Gene: MME —**ASSOCIATED_WITH**→ Disease: Heart failure [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)

Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.

No mapped literature claim supports a critical biological bridge; score is capped below moderate support.

Missing evidence: Direct biological bridge for Target: MMP8 (P22894) → Pathway: Neutrophil degranulation (R-HSA-6798695) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Pathway: Neutrophil degranulation (R-HSA-6798695) → Gene: MME is not supported by a mapped claim.

Candidate Mechanism 5: Mechanism 5: MMP1 (P03956) via Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) [WEAK_SPECULATIVE] [DIR: UNKNOWN]

Drug: Doxycycline → Target: MMP1 (P03956) → Reaction: Expression of STAT3-upregulated extracellular proteins (R-HSA-6789615) → Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) → Gene: CDKN1A → Disease: Heart failure

- Drug: Doxycycline —**INHIBITOR**→ Target: MMP1 (P03956) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)

Status: DATABASE SUPPORTED; Directionality: SUPPORTED; Source: ChEMBL

- Target: MMP1 (P03956) —**ENTITY_SET_MEMBER_OF**→ Reaction: Expression of STAT3-upregulated extracellular proteins (R-HSA-6789615) [\[Open Reactome Reaction\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome

- Reaction: Expression of STAT3-upregulated extracellular proteins (R-HSA-6789615) —**PART_OF**→ Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) [\[Open Reactome Pathway\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome

- Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) —**CONTAINS_ASSOCIATED_GENE**→ Gene: CDKN1A [\[Open Reactome Pathway\]](#) [\[Open Targets Target\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome + Open Targets/DisGeNET

- Gene: CDKN1A —**ASSOCIATED_WITH**→ Disease: Heart failure [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)

Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.

No mapped literature claim supports a critical biological bridge; score is capped below moderate support.

Missing evidence: Direct biological bridge for Target: MMP1 (P03956) → Reaction: Expression of STAT3-upregulated extracellular proteins (R-HSA-6789615) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Reaction: Expression of STAT3-upregulated extracellular proteins (R-HSA-6789615) → Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Pathway: Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807) → Gene: CDKN1A is not supported by a mapped claim.

Reaction-Enriched Mechanistic Evidence (Phase 3)

Reactome reaction evidence describes the target's molecular role within the reaction. It does not by itself establish therapeutic activation or inhibition. Direction defaults to UNKNOWN.

Target	Reaction / Event	Role	Pathway	Direction	Source
MMP1 (P03956)	Basigin binds Matrix metalloproteinase-1	COMPLEX COMPONENT	Basigin Interactions (R-HSA-210991)	DIRECTION_UNKNOWN	Reactome
MMP1 (P03956)	Expression of STAT3-upregulated extracellular matrix	ENTITY SET MEMBER	Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807)	DIRECTION_UNKNOWN	Reactome

Biomedical Data Sources Accessed

Source Name	Status	Direct Source Portal
ChEMBL	SUCCESS	[Open ChEMBL]
UniProt	SUCCESS	[Open UniProt]
Reactome	SUCCESS	[Open Reactome]
Open Targets	SUCCESS	[Open Open Targets]
DisGeNET	SUCCESS	[Open DisGeNET]
PubMed	SUCCESS	[Open PubMed]
Europe PMC	SUCCESS	[Open Europe PMC]
ClinicalTrials.gov	SUCCESS	[Open ClinicalTrials.gov]

Recommendation Rule Engine Output

- Rule 1 (PROMISING): SS = 0.987 (≥ 0.40), MS = 0.409 (≥ 0.40), RS = 0.058 (≤ 0.39). Safety grade: B.
- Evidence signals:
 - [PASS] Literature support: SS = 0.987 (HIGH) from 62 records
 - [PASS] Mechanistic plausibility: MS = 0.409 (LOW), 10 pathway(s)
 - [PASS] Safety/Risk acceptable: RS = 0.058 (LOW)
 - [PASS] Evidence consistency: No contradictions
 - [PASS] Human clinical data: Available

Data Gaps Identified

- No prior knowledge entries found for this drug-disease pair. This appears to be a genuinely novel repurposing hypothesis (−0.05 confidence penalty).

Evidence Synthesis & Dimensional Assessment

1. Mechanistic Assessment

Mechanistic Assessment: LOW (Score = 0.409).

Literature Grounding: MODERATE.

Candidate Biological Mechanisms Discovered: 5.

Primary Mechanistic Chain: Drug: Doxycycline → Target: MMP1 (MMP1 (Interstitial collagenase) [UniProt: P03956]) [INHIBITOR via ChEMBL] → Pathway: Basigin interactions (R-HSA-210991) [PARTICIPATES_IN via Reactome] → Gene: ATP1B1 [CONTAINS_ASSOCIATED_GENE via Reactome + Open Targets/DisGeNET] → Disease: Heart failure [ASSOCIATED_WITH via Open Targets / DisGeNET].

Rationale: Mechanistic Score (MS) derived from candidate mechanism quality. Best candidate support level: WEAK_SPECULATIVE. Discovered 5 candidate biological mechanism(s). Literature grounding: MODERATE (7 claim(s)). Final MS = 0.409 (LOW).

2. Clinical Evidence Assessment

Clinical Evidence Assessment: HIGH Literature Support (Score = 0.987).

Clinical Trial Status: RETRIEVED.

Clinical Trial Findings: 4 trial(s) evaluated, 0 clinically significant failure(s).

Literature Support Rationale: Support Score (SS) from 62 evidence records.

Breakdown: Cell-line (in vitro): 5 records (avg ERW: 0.30); Reviews/Observational: 55 records (avg ERW: 0.62); Extracted claims: 2

Formula: $SS = 1 - \exp(-0.12 \times \text{quality_weighted_sum})$. Quality-weighted sum = 36.15 (raw ERW sum would be 37.90, capped to prevent inflation from low-quality records).

Final SS = 0.987 (HIGH).

3. Safety & Risk Assessment

Safety & Risk Assessment: Grade B — Acceptable — minor concerns, generally safe (Risk Score = 0.058, LOW).

Boxed Warning: NO.

Risk Factors & Signal Analysis: Risk Score (RS) computed from clinically significant safety signals.

Trial analysis: 0 clinically significant failure(s); 0 administrative/external termination(s) excluded.

Risk factors: Safety grade: B (grade penalty: 0.2).

Formula: $RS = 1 - \exp(-0.30 \times \text{raw_risk_burden})$. Raw burden = 0.20. Final RS = 0.058 (LOW).

4. Final Synthesis

Synthesis & Recommendation: PROMISING.

Regulatory Status: NONE (Term: N/A).

1. Mechanistic Plausibility: LOW (0.409)

2. Literature/Clinical Support: HIGH (0.987)

3. Risk & Safety Profile: LOW (0.058, Grade B)

Conclusion Summary: Rule 1 (PROMISING): SS = 0.987 (≥ 0.40), MS = 0.409 (≥ 0.40), RS = 0.058 (≤ 0.39). Safety grade: B.

Confidence Narrative

Overall confidence is estimated at 95% after accounting for data gaps (total penalty: -5%). Evidence quality: The analysis is based on 62 retrieved records representing moderate-quality mixed evidence. ERW values are capped per evidence tier to prevent inflation from low-quality records (e.g., reviews and

hypotheses). Safety assessment: Grade B — Acceptable — minor concerns, generally safe. No boxed warning detected. Safety grade scale: A = Excellent | B = Acceptable | C = Caution | D = Unsafe. Prior knowledge: No ChEMBL indication data available and no cache entries found for Doxycycline → Heart failure. Treating as a novel repurposing hypothesis. Evidence evaluated on its own merits.